

ASD Screening Game Design Document

Game Overview

Parameter	Specification
Platform	Godot 2D
Target Age	3-10+ years
Total Duration	75-90 minutes
Purpose	Screen for ASD indicators through gameplay-based behavioral assessment

Core Data Collection Metrics

- Response time (milliseconds)
- Number of attempts before correct response
- Mouse movement patterns
- Sequence of selections
- Task completion consistency
- Help request frequency

CHAPTER 1: Welcome to My World

Duration: 5-7 minutes | **Focus:** Baseline & Response to Name

Level 1.1: Avatar Creation & Name Response

Tasks:

- Avatar customization (hair, clothes, colors)
- Character calls child's name 3 times at 5-second intervals
- Child must respond by clicking character

Scoring:

Response Time	Score
< 2 seconds	0 pts

Response Time	Score
2-5 seconds	1 pt
> 5 seconds or no response	2 pts
Multiple attempts needed	+1 pt each

ASD Indicator: M-CHAT #10 - Response to name

Level 1.2: Following the Guide

Tasks:

- Animal guide points to 5 objects (tree, house, flower, sun, cloud)
- Child clicks where guide points

Scoring:

Behavior	Score
All 5 correct on first try	0 pts
Needs verbal prompt	1 pt each
Clicks pointing finger vs target	2 pts each
Random clicking/ignoring	3 pts

ASD Indicator: Joint attention deficit

CHAPTER 2: Emotion Island

Duration: 10-15 minutes | Focus: Emotion Recognition & Expression

Level 2.1: Emotion Matching Garden

Tasks:

- 4 flower beds (Happy, Sad, Angry, Scared)
- Phase 1: Match 12 faces (3 per emotion) - child, adult, animal faces
- Phase 2: Match 8 contextual scenarios to emotions

Scoring:

Accuracy	Score
> 90%	0 pts
70-90%	1 pt
50-70%	2 pts
< 50%	3 pts
Negative emotions < 80% accurate	+2 pts
Fear/Sad confusion rate > 25%	+2 pts

ASD Indicator: Difficulty with negative emotion recognition

Level 2.2: Emotion Expression Mirror

Tasks:

- Show emotion label + voice: "Show me a HAPPY face!"
- Select from 6 face options (1 correct, 5 distractors)
- 16 trials (4 per emotion)

Scoring:

Selection	Score
Correct emotion	0 pts
Neutral face when emotion requested	2 pts
Opposite emotion	3 pts
Intensity error	1 pt

Level 2.3: Emotion Regulation Scenarios

Scenarios (6 total):

- Ice cream drop (sadness)
- Friend takes toy (anger)
- Loud noise (sensory)
- Lost item (frustration)
- Broken toy (disappointment)
- Social conflict (peer interaction)

Response Options Per Scenario:

- Socially appropriate (0 pts)
- Aggressive/inappropriate (3 pts)
- Avoidant (2 pts)
- Decision time > 15s (+1 pt)

CHAPTER 3: Friend's House Visit

Duration: 12-15 minutes | Focus: Social Communication

Level 3.1: Greeting Sequence

Task Flow:

Step	Options	Scoring
Door knock	"Hello!", silence, "Can I play?", "Where's friend?"	No greeting: 3 pts, Inappropriate: 2 pts, Delayed appropriate: 1 pt
Friend waves/smiles	Wave back, smile back, both, nothing	No reciprocation: 2 pts
Eye contact simulation	Click friend's face/eyes within 5s	No click or > 4s delay: 2 pts

ASD Indicator: M-CHAT #11 - Social reciprocity

Level 3.2: Conversation Turn-Taking

Structure: 8 conversational exchanges with 3-4 response options each

Example Exchange:

- Friend: "I got a new puzzle! Want to help?"
- Options: "Yes, fun!" (appropriate) | "I only like cars" (rigid) | silence | dinosaur facts (off-topic)

Scoring:

Response Type	Score
Off-topic (restricted interest)	3 pts each
Overly literal when social expected	2 pts each
No response within 10s	2 pts
Pattern: always factual over emotional	+5 pts

Level 3.3: Sharing & Joint Attention

Tasks:

- Friend points to discovery (butterfly, rock, marble) - child must click object
- Child finds items - choose to show friend, keep playing alone, info-dump, pocket it

Scenarios: 5 discovery events

Scoring:

Behavior	Score
Doesn't attend to friend's discoveries	2 pts each
Doesn't show own discoveries	2 pts each
Excessive factual detail vs sharing joy	1 pt each

ASD Indicators: M-CHAT #9 (showing to share), M-CHAT #16 (joint attention)

CHAPTER 4: Daily Routines Village

Duration: 10-12 minutes | **Focus:** Executive Function & Cognitive Flexibility

Level 4.1: Morning Routine Sequence

Task: Drag-drop 6 activities in order

1. Wake up
2. Brush teeth
3. Get dressed
4. Eat breakfast
5. Pack bag
6. Put on shoes

Disruption: "No clean shirt! What do you do?"

- Ask parent (flexible - 0 pts)
- Wear dirty shirt (rigid - 3 pts)
- Meltdown (inflexibility - 4 pts)
- Yesterday's shirt (problem-solving - 0 pts)

Scoring:

Metric	Score
Sequence errors	1 pt each
Cannot complete after 3 attempts	3 pts
Rigid response to disruption	3 pts
Response time to disruption > 10s	1 pt

Level 4.2: Playground Equipment Choice

Structure:

- 5 activities: swing, slide, sandbox, monkey bars, seesaw
- Select → play (3 clicks) → "Try something new!" → repeat 4 times

Scoring:

Behavior	Score
Same activity 3+ times	3 pts
Transition delay > 8s each	2 pts
Exact same sequence every loop	2 pts

Level 4.3: Unexpected Events Response

Scenarios (4 total):

Situation	Options	Scoring
Path blocked	Detour, wait, upset, ask help	Flexible: 0 pts, Rigid: 3 pts, Distress: 2 pts, Refusal: 3 pts
Library closed	Tomorrow, other activity, cry, try anyway	Same scoring pattern
Food unavailable	Try new, order nothing, leave upset, ask similar	Same scoring pattern
Rain cancels outdoor	Indoor alternative, insist outdoor, refuse all, suggest plan	Same scoring pattern

Response time > 12s: +1 pt per scenario

CHAPTER 5: Pretend Play Theater

Duration: 8-10 minutes | **Focus:** Symbolic Play & Imagination

Level 5.1: Pretend Play Recognition

Scenarios (5 animations):

Character Action	Question	Options
Banana to ear	"What are they doing?"	Eating, talking on phone (pretend), being silly, holding
Stir empty pot	"What are they doing?"	Cleaning, cooking (pretend), playing, nothing
Feed teddy with empty spoon	"What are they doing?"	Feeding bear (pretend), playing, holding spoon, moving
Push block making car sounds	"What are they doing?"	Pushing, playing car (pretend), noise, moving
Pretend drink empty cup	"What are they doing?"	Holding, drinking (pretend), playing, looking

Scoring:

Response Type	Score
Literal when pretend obvious	2 pts each
"Don't know"/no response	1 pt each
All 5 literal	+3 pts pattern flag

ASD Indicator: M-CHAT #3 - Pretend play deficit

Level 5.2: Create Your Own Pretend Play

Prompts & Objects (4 scenarios):

Prompt	Objects Available	Target Symbolic Use
"Tea party!"	Blocks, stick, leaves, rocks, toy cup, paper	Leaves=food, rocks=plates
"Be superheroes!"	Blanket, box, pillow, toy, stick	Blanket=cape, box=hideout
"Go on trip!"	Chair, basket, cardboard, rope, bowl	Chair=car, basket=suitcase

Prompt	Objects Available	Target Symbolic Use
"Build house!"	Blocks, fabric, sticks, box	Various construction

Scoring:

Behavior	Score
No symbolic selections (only literal)	4 pts per scenario
Some symbolic but rigid	2 pts
Refuses to engage/selects nothing	3 pts per scenario
Response time > 15s	+1 pt

CHAPTER 6: Sensory Garden

Duration: 6-8 minutes | Focus: Sensory Processing

Level 6.1: Sound Sensitivity

8 Sounds Presented:

Sound	Type	Volume Level
Birds chirping	Pleasant	Soft
Water fountain	Pleasant	Medium
Children laughing	Social	Medium-loud
Vacuum cleaner	Mechanical	Loud
Dog barking	Sudden	Loud
Thunder	Sudden	Very loud
Baby crying	High-pitched	Loud
Traffic	Constant	Loud

Response Options: Happy face (0 pts), Neutral (0 pts), Worried (1 pt), Upset (2 pts), Cover ears (+1 pt), Leave immediately (+2 pts)

Scoring:

Metric	Score
4+ sounds rated distressing	3 pts
Vacuum/mechanical distress	2 pts (specific flag)
Covers ears 3+ sounds	2 pts
Leaves immediately 2+ times	3 pts

ASD Indicator: M-CHAT #12 - Everyday noise sensitivity

Level 6.2: Visual Sensitivity Patterns

6 Visual Rooms:

Room	Content	Type
Calm painting	Simple, soft colors	Static pleasant
Bright rainbow	Colorful	Static bright
Flickering lights	Subtle flashing	Animated
Spinning pinwheel	Motion	Animated
Crowded scene	Many objects	Complex static
Striped patterns	High contrast repetitive	Static complex

Actions: Walk quickly past (avoidance), Stop and look (engagement), Rate comfort

Scoring:

Behavior	Score
Avoids 3+ rooms	2 pts
Distress at flickering/motion	2 pts
Excessive time in repetitive patterns (> 20s when others < 10s)	1 pt
Distress at crowded scene	1 pt

Level 6.3: Texture Preferences

8 Textures Described (visual + audio):

Texture	Category	Common ASD Response
Soft fluffy cotton	Pleasant	Typically positive

Texture	Category	Common ASD Response
Smooth cool glass	Neutral	Typically neutral
Bumpy rough rock	Unpleasant	May avoid
Squishy wet clay	Wet/sensory	Often aversive
Sticky honey	Wet/sticky	Often aversive
Scratchy sandpaper	Unpleasant	May avoid
Silky ribbon	Pleasant	Typically positive
Slimy jello	Wet/texture	Often aversive

Response Options: Love to touch (0 pts), Okay (0 pts), Don't like (1 pt), Never touch (2 pts), Won't try (3 pts)

Scoring:

Metric	Score
4+ rated don't like or worse	2 pts
Wet textures (clay, honey, jello) all aversive	2 pts
Refuses to try 2+	2 pts

CHAPTER 7: Pattern Detective

Duration: 8-10 minutes | **Focus:** Restricted/Repetitive Behaviors

Level 7.1: Pattern Completion

Pattern Sequence:

- 1. Simple AB: Red, blue, red, blue, ?, ?
- 2. ABC: Circle, square, triangle, circle, square, ?, ?
- 3. Complex: Red circle, red square, blue circle, blue square, ?, ?, ?, ?
- 4. Forced error: Game places wrong item - "Fix it or leave it?"
- 5. Pattern change: "Let's try a new pattern!"

Scoring:

Behavior	Score
Pattern errors	0 pts (cognitive, not ASD)
Strong distress at error ("must fix + upset")	3 pts
Refuses new pattern	3 pts
Delay > 10s to start new pattern	2 pts
Keeps returning to first pattern	2 pts

Level 7.2: Repetitive Behavior Recognition

Free Play Observation Tracking:

- Same action repeated 8+ times consecutively (2 pts)
- Spontaneous organizing/lining up items (2 pts)
- Distress when organized items disrupted (2 pts)

Level 7.3: Special Interest Intensity

8 Topic Categories:

- Dinosaurs
- Trains
- Space/planets
- Animals
- Vehicles
- Numbers/math
- Weather
- Maps/geography

Structure:

- Free choice → read facts → "Try different book!" → repeat (4 opportunities)
- Track: topic selection, engagement depth (fact clicks), transition resistance

Scoring:

Behavior	Score
Same topic 3+ times	3 pts
Reads 15+ facts single topic	2 pts (unusual depth)
Transition resistance (delay > 8s)	2 pts each

Behavior	Score
Returns to same topic after forced change	2 pts

CHAPTER 8: Copy Cat Challenge

Duration: 6-8 minutes | **Focus:** Imitation Skills

Level 8.1: Simple Action Imitation

12 Actions (select matching from 4 options):

Category	Actions
Facial	Wave hello, clap hands, touch nose, stick out tongue
Body	Jump up/down, spin around, stomp feet, arms up high
Object Use	Pretend drink cup, pretend comb hair, pretend throw ball, pretend read book

Scoring:

Error Type	Score
Facial imitation	2 pts each
Body imitation	1 pt each
Object use	1 pt each
No response within 8s	2 pts each
6+ total errors	+3 pts (significant deficit)

ASD Indicator: M-CHAT #15 - Imitation deficit

Level 8.2: Sequential Imitation

6 Sequences:

Sequence	Complexity
Clap → Wave	2 actions
Jump → Spin	2 actions
Touch head → Touch nose	2 actions

Sequence	Complexity
Wave → Clap → Jump	3 actions
Touch nose → Stomp → Clap	3 actions
Spin → Wave → Touch head	3 actions

Scoring:

Metric	Score
2-action sequence errors	1 pt each
3-action sequence errors	2 pts each
Same action repeated (perseveration)	+2 pts
Cannot complete any 3-action	+3 pts

CHAPTER 9: Assessment Summary

Duration: 8-10 minutes | **Focus:** Consistency & Generalization

Structure:

- 20 randomly selected tasks from all previous chapters
- No new mechanics
- Tests: consistency across contexts, skill maintenance, pattern emergence

SCORING ALGORITHM

Domain Breakdown

Domain	Chapters	Max Points	Risk Levels
1. Social Communication	2, 3, 8	100	Low: 0-20, Medium: 21-45, High: 46+
2. Restricted/Repetitive	4, 7	70	Low: 0-15, Medium: 16-30, High: 31+
3. Sensory Processing	6	30	Low: 0-8, Medium: 9-15, High: 16+
4. Pretend Play	5	40	Low: 0-10, Medium: 11-20, High: 21+

Combined Score Formula

Total = (Domain 1 × 0.4) + (Domain 2 × 0.3) + (Domain 3 × 0.15) + (Domain 4 × 0.15)

Maximum: 100 points

Overall Risk Classification

Score Range	Risk Level	Recommendation
0-25	Low	Minimal indicators, typical development likely
26-45	Medium	Some indicators, monitor and rescreen in 3-6 months
46-65	High	Significant indicators, clinical evaluation recommended (ADOS-2)
66+	Very High	Strong indicators across domains, urgent clinical referral

Red Flag Multipliers

Pattern	Multiplier
Negative emotion recognition < 50%	×1.2
Complete absence pretend play recognition	×1.3
Extreme sensory sensitivities (4+ distressing sounds)	×1.15
Rigid pattern behavior + distress at changes	×1.2
Poor imitation across all modalities	×1.25

TECHNICAL IMPLEMENTATION

Data Structure

Player Data Singleton:

- Session ID
- Age
- Start timestamp
- Response dictionary (chapter/level/task/response/time/attempts)
- Score dictionary
- Timing data (millisecond precision)
- Mouse movement array (x, y coordinates every 100ms)

Key Systems

Response Time Tracker:

- Start: task presentation
- End: interaction
- Precision: milliseconds

Attempt Counter:

- Reset: new task
- Increment: incorrect selection
- Log: selection sequence

Save/Load:

- Format: JSON
- Features: resume capability, researcher dashboard export
- Anonymization: GDPR compliant

VALIDATION REQUIREMENTS

Three-Phase Validation

Phase	Participants	Purpose	Duration
1. Normative Data	100+ neurotypical children ages 3-10	Establish baseline scores per age group	8-12 weeks
2. Clinical Validation	50+ diagnosed ASD children with ADOS-2 scores	Calculate sensitivity/specificity	8-12 weeks
3. Iteration	Based on false positive/negative rates	Adjust difficulty, scoring, flags	4-8 weeks

DEVELOPMENT TIMELINE

Phase	Duration	Details
Asset Creation	4-6 weeks	Character sprites (4 emotions × 3 intensities), 27 backgrounds, UI elements, sound effects, animations

Phase	Duration	Details
Core Mechanics	3-4 weeks	Response tracking, attempt counting, mouse movement, save system
Chapter Implementation	8-10 weeks	~1 week per chapter average
Data System	2 weeks	JSON export, researcher dashboard, anonymization
Testing/Debugging	3-4 weeks	Quality assurance, bug fixes
Validation Studies	12+ weeks	External clinical validation
Total to Beta	5-6 months	Playable, validated prototype

ASSET REQUIREMENTS

Visual Assets

Category	Quantity	Specifications
Character sprites	12	4 emotions × 3 intensities
Background scenes	27	9 chapters × avg 3 scenes
UI elements	50+	Buttons, response options, icons, progress indicators
Object images	200+	Vocabulary items, toys, daily objects

Audio Assets

Category	Quantity	Specifications
Sensory sounds	8	Birds, fountain, vacuum, dog, thunder, baby, traffic
Feedback sounds	10	Positive, negative, neutral, transitions
Voice instructions	100+	Task prompts, questions, narration
Ambient music	5	Background tracks per chapter theme

Animation Assets

Category	Quantity	Purpose
Imitation actions	12	Wave, clap, jump, spin, facial expressions, object use
Pretend play scenarios	5	Banana phone, empty pot stirring, teddy feeding, block car, empty cup
Emotion regulation stories	6	Ice cream drop, toy taking, loud noise, lost item, broken toy, conflict

SUCCESS METRICS

Criterion	Target
Standardization	100% consistent stimuli presentation
Engagement	> 85% completion rate across age groups
Simplicity	Ages 3-4 can complete with < 10% help requests
Data Quality	< 2% missing data points per session
Clinical Validity	Correlation with ADOS-2 > 0.75
Sensitivity	> 80% true positive rate
Specificity	> 75% true negative rate

References

[1]

- **Title:** Autism Screening Using a Video Game Based on Emotions
- **Authors:** Atefeh Irani, Hadi Moradi, Leila Kashani Vahid
- **URL:** <https://ieeexplore.ieee.org/document/8712053>
- **Citation:** A. Irani, H. Moradi and L. K. Vahid, "Autism Screening Using a Video Game Based on Emotions," 2018 2nd National and 1st International Digital Games Research Conference: Trends, Technologies, and Applications (DGRC), Tehran, Iran, 2018, pp. 40-45, doi: 10.1109/DGRC.2018.8712053.

[2]

- **Title:** Modified Checklist for Autism in Toddlers, Revised, with Follow-Up (M-CHAT-R/F)
- **Authors:** Diana L. Robins, Deborah Fein, Marianne Barton
- **PDF-URL (public):** https://drexel.edu/~media/Files/autismInstitute/EDI/M-CHAT-R_F.ashx

[3]

- **Title:** Serious Game for Children with Autism to Learn Vocabulary - An Experimental Evaluation
- **DOI:** <https://doi.org/10.1080/10447318.2017.1420006>
- **Journal:** International Journal of Human-Computer Interaction
- **Authors:** Khowaja, Kamran, Salim, Siti Salwah
- **PDF-URL (mendeley):** <https://www.mendeley.com/reference-manager/reader/3a04119b-da8d-38d1-a853-ada717229a36/6a32e3cb-91f0-1691-6864-64d09e5bcbbb>
- **Citation Key:** Khowaja2019

[4]

- **Title:** The Accuracy of the ADOS-2 in Identifying Autism among Adults with Complex Psychiatric Conditions
- **URL:** <https://pmc.ncbi.nlm.nih.gov/articles/PMC5813679/>
- **Authors:** Brenna B. Maddox, Edward S. Brodtkin, Monica E. Calkins, Kathleen Shea, Katherine Mullan, Jack Hostager1, David S. Mandell, and Judith S. Miller
- **PDF-URL (public):** <https://pmc.ncbi.nlm.nih.gov/articles/PMC5813679/pdf/nihms939651.pdf>

[5]

- **ADOS-2:** <https://www.actionbehavior.com/screener/ados-2>
-